

Write a program to implement Linear Regression (LR) algorithm in python

CODE:

```
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
X = np.array([1, 2, 3, 4, 5, 6, 7, 8]).reshape(-1, 1)
y = np.array([2, 4, 5, 8, 10, 12, 14, 16])
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("Linear Regression Results")
print("-----")
print("Slope:", model.coef_[0])
print("Intercept:", model.intercept_)

print("\nActual Values:", y_test)
print("Predicted Values:", y_pred)
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)
print("\nMean Squared Error:", mse)
print("R2 Score:", r2)
new_value = np.array([[9]])
prediction = model.predict(new_value)
```

```
print("\nPrediction for X = 9:", prediction[0])
```

OUTPUT:

```
... Linear Regression Results
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Slope: 2.0499999999999994
Intercept: -0.3999999999999986

Actual Values: [ 4 12]
Predicted Values: [ 3.7 11.9]

Mean Squared Error: 0.050000000000000009
R2 Score: 0.996875

Prediction for X = 9: 18.049999999999997
```
